

## Training Configuration and Summary:

Dataset used: rssi\_odom\_dataset\_1204.parquet

### LSTM Multi-Step Training Report

Report Generated: 2025-05-06 14:38:56

Training Start: 2025-05-06 13-32-50

Training End: 2025-05-06 13-36-47

Training Duration: 3.94 minutes

#### Best Hyperparameters:

Number of Layers: 2

Activation: tanh

Optimizer: rmsprop

Dropout Rate: 0.20

MAE: 1.5855

MSE: 3.9996

RMSE: 1.9999

R2 Score: 0.7656

#### Multi-Step Forecast:

Forecast Steps: 20

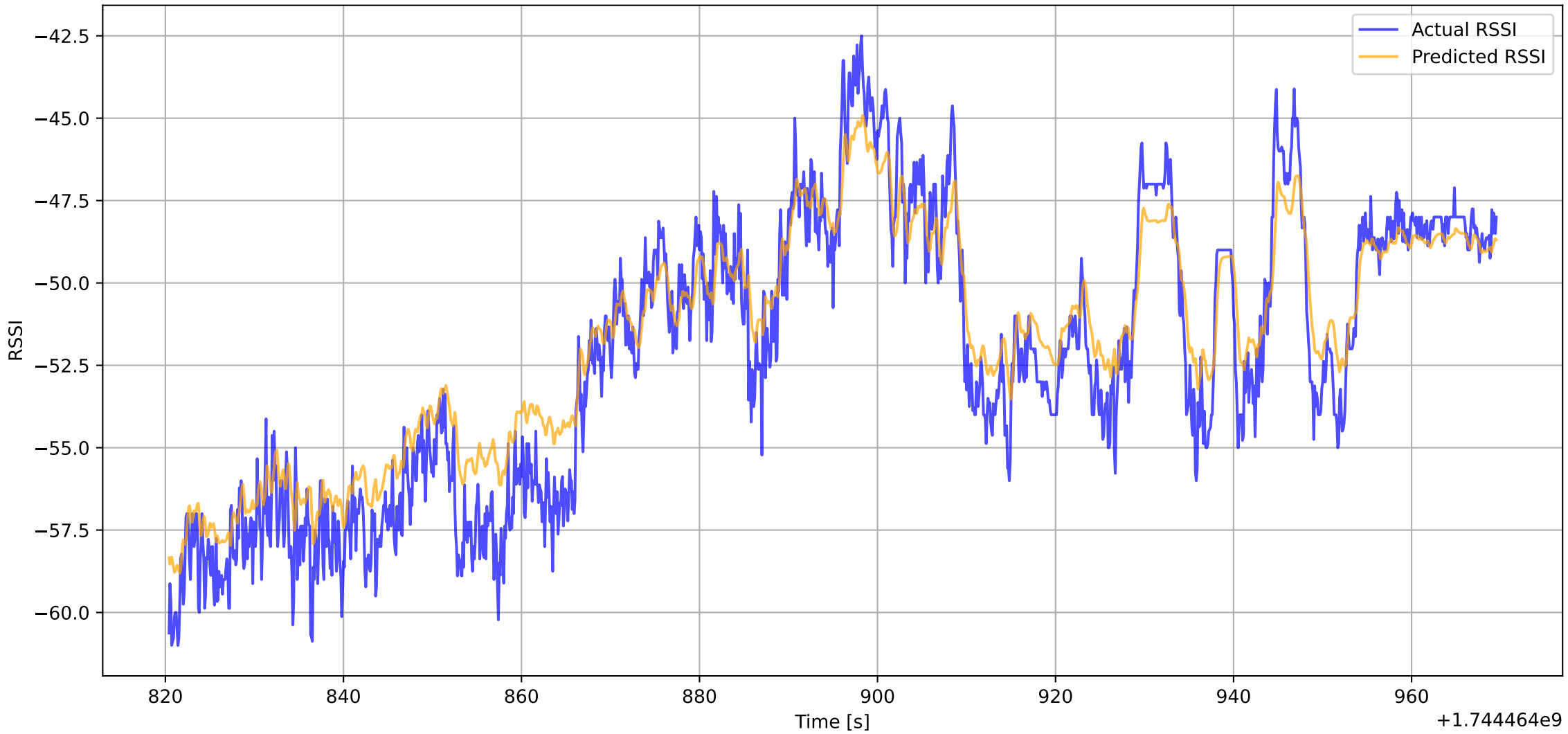
MAE: 0.0432

MSE: 0.0028

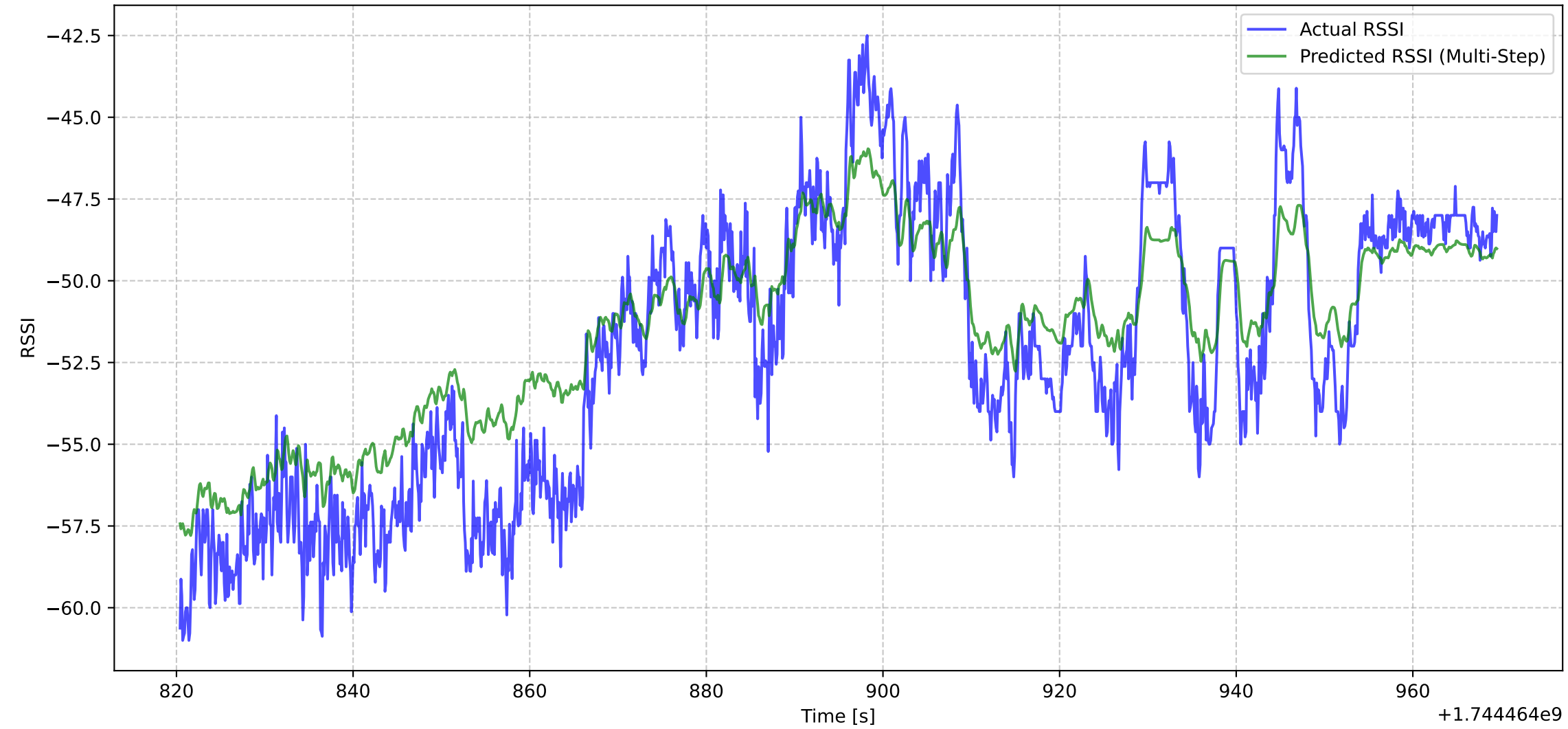
RMSE: 0.0532

R2 Score: 0.8798

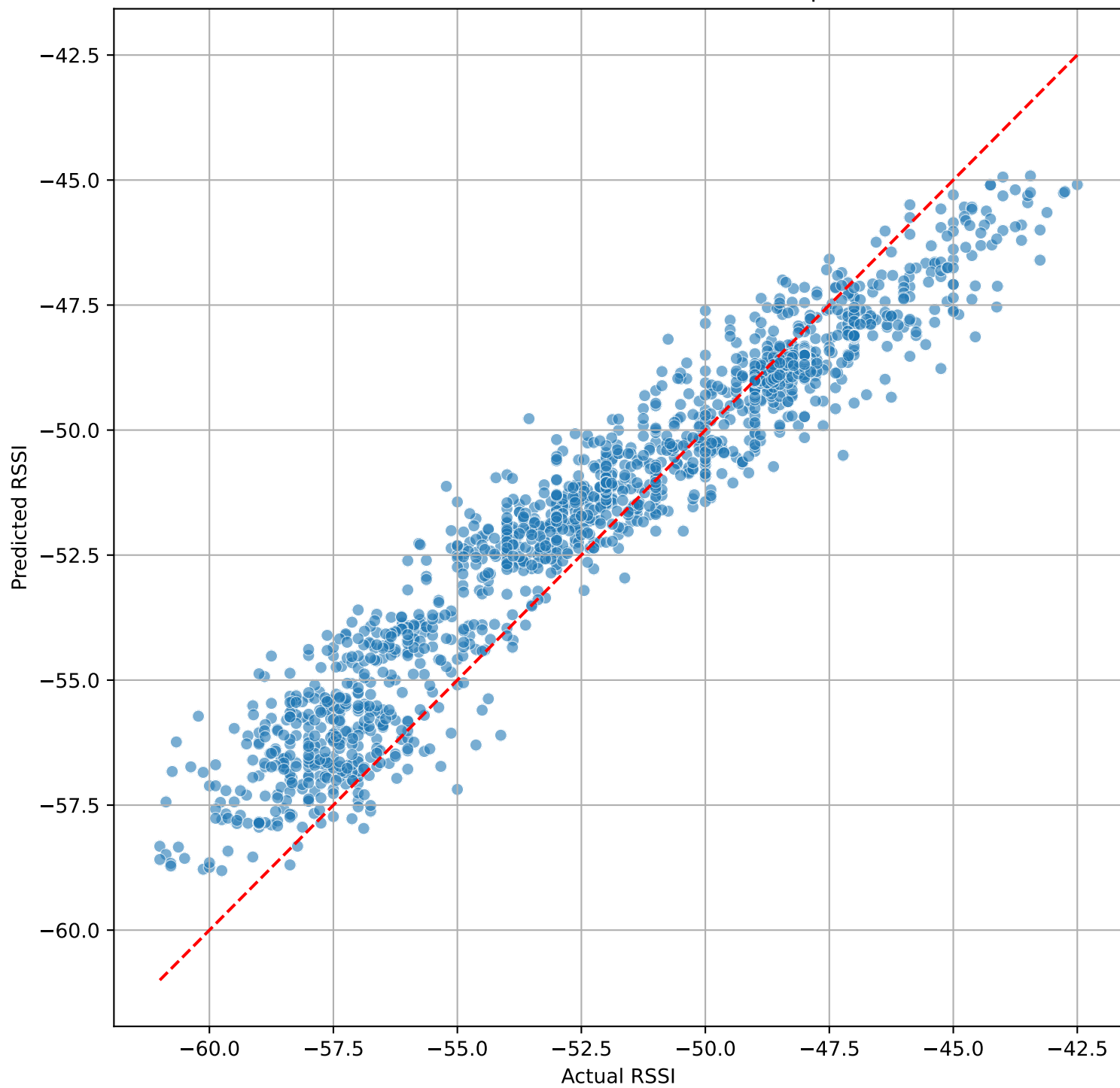
Actual vs Predicted RSSI (First Step)



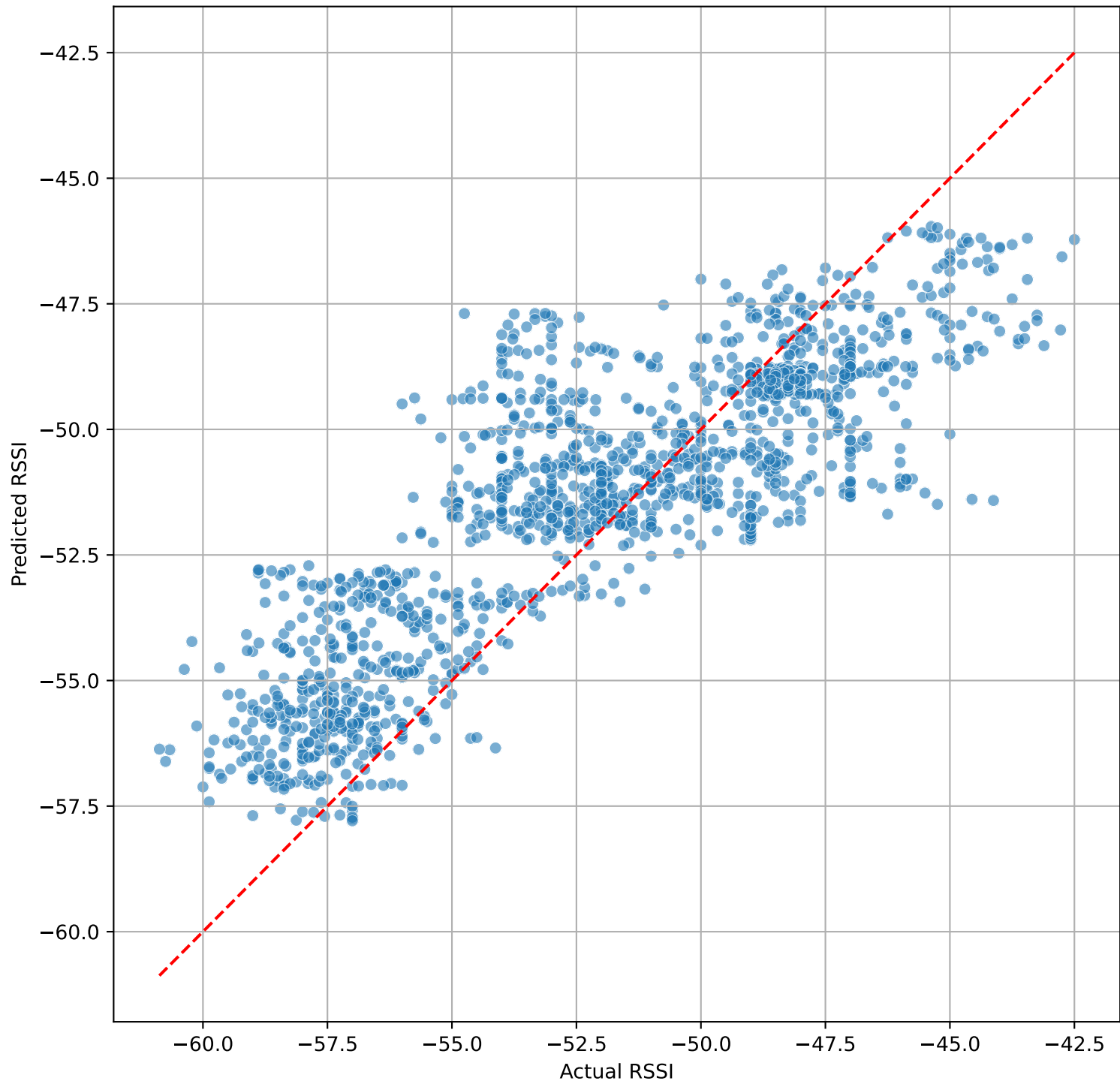
Actual vs Predicted RSSI (Multi-Step (step 20))



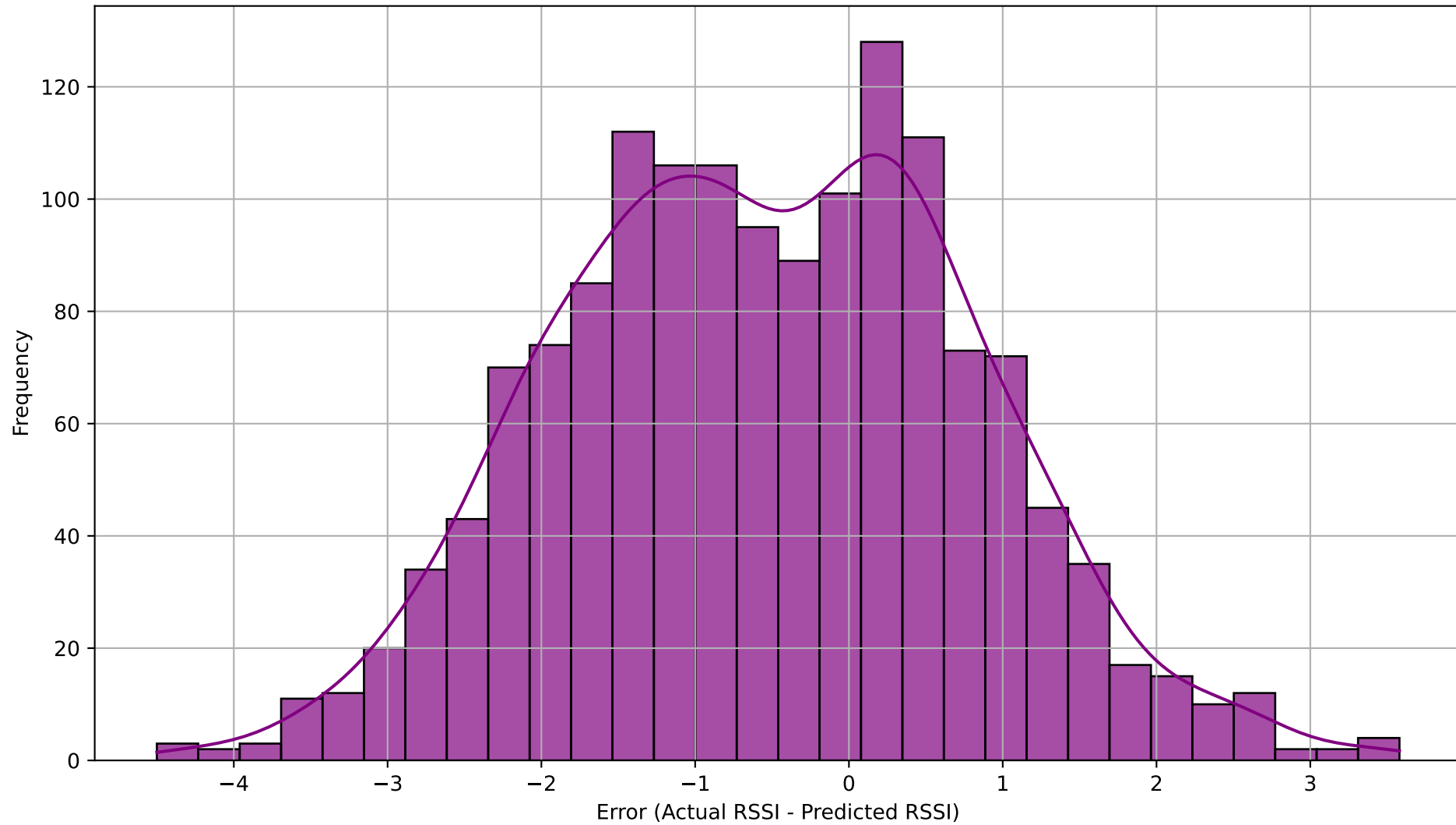
Actual vs Predicted RSSI (First Step)



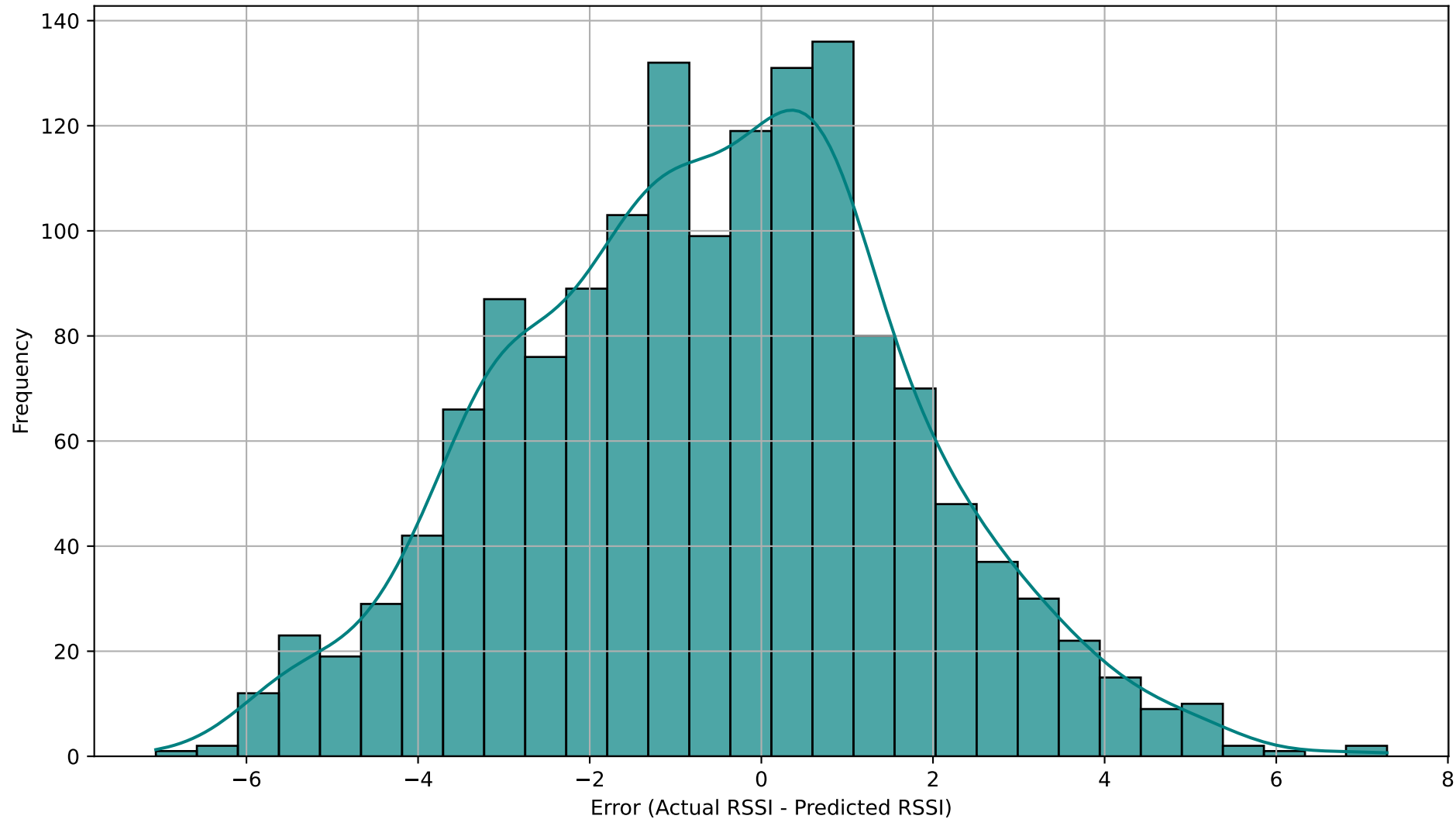
Actual vs Predicted RSSI (Last Step - step 20)



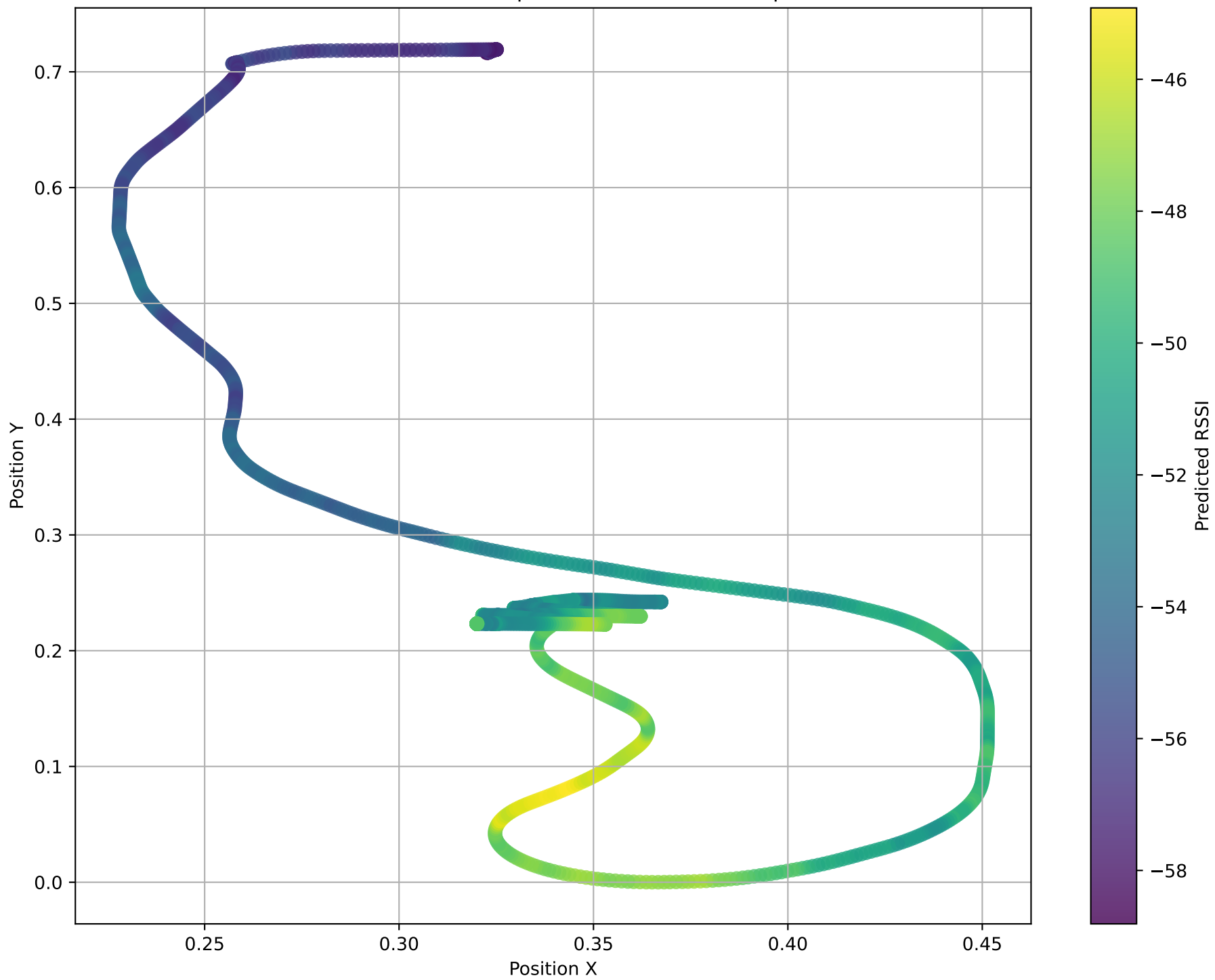
Distribution of Prediction Errors (First Step)



Distribution of Prediction Errors (Last Step - step 20)

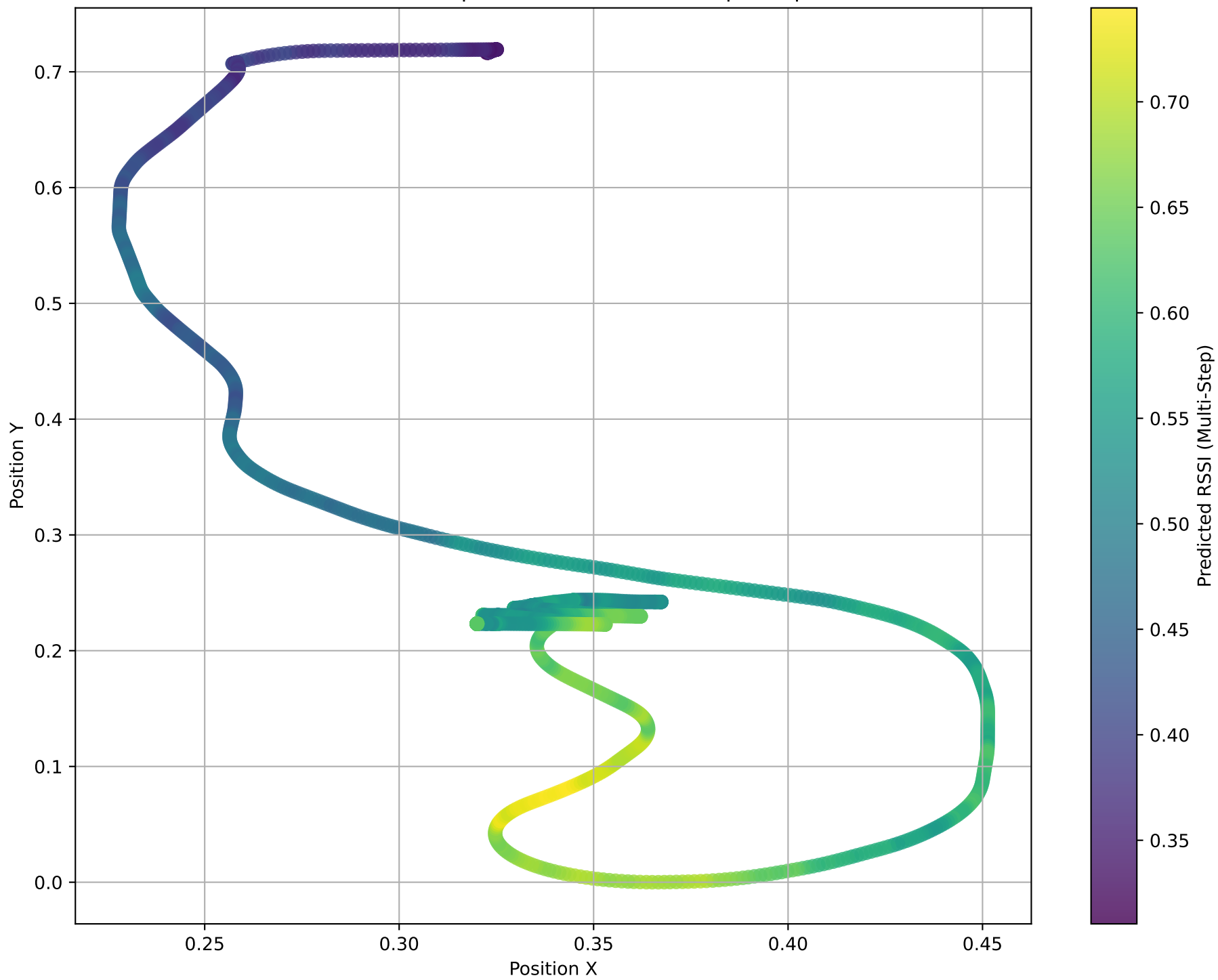


Predicted RSSI in Spatial Context (First Step)

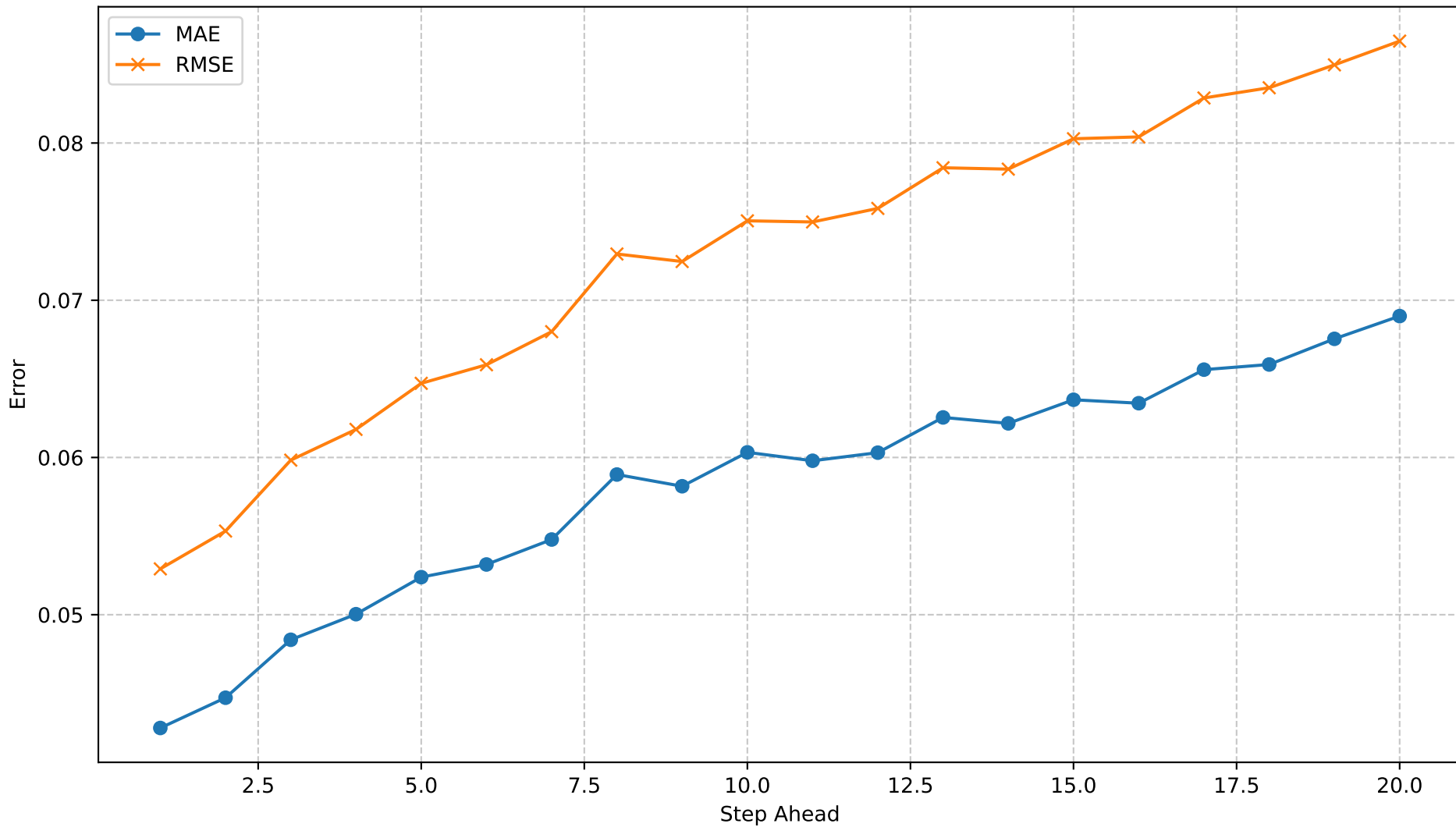




Predicted RSSI in Spatial Context (Multi-Step - step 20)



Multi-Step Forecast Error by Step



#### Resumo Automático:

- Forecast Steps configurados: 20
- Passos com MAE  $\leq 2.0$ : 20 passo(s)
- MAE no último passo: 0.0690

[ V ] O modelo mantém boa acurácia durante quase toda a sequência de previsões.